Python(22AIE205) Programming Lab

Weekly Lab Report-1

LAB 1: INPUT, OUTPUT AND ARITHMETIC OPERATIONS.

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Experiment Date:	

Experiment Record:

Date:	14-07-2026
Experiment Name:	INPUT, OUTPUT AND ARITHMETIC OPERATIONS
Aim:	To practise keyboard input, formatted output, type conversion, arithmetic operators, floor division, modulus, and exponentiation using Python IDLE.
Software Required:	Python 3.x and Python IDLE

Questions Given:

The following questions were assigned for Lab 1. The detailed programs and sample outputs are presented in the next section.

Question 1: Printing Personal Details - Write a Python program to accept and display the student name, registration number, age, department, university name, and city. Also print one sentence using an f-string.

Question 2: Sum of Two Numbers - Accept two integers from the user and display their sum.

Question 3: Basic Arithmetic Operations - Accept two numbers and display their sum, difference, product, quotient, floor quotient, remainder, and power.

Question 4: Square and Cube - Accept one number and calculate its square and cube using the exponentiation operator (**).

Question 5: Simple Interest - Accept principal amount, annual rate of interest, and time period. Calculate the simple interest and total amount.

Question 6: Area of a Triangle - Accept the base and height of a triangle and calculate its area.

Question 7: Kilometre Conversion - Accept a distance in kilometres and convert it into metres and centimetres.

Question 8: Salary Calculation - Accept the basic salary, house rent allowance (HRA), and travel allowance. Calculate and display the gross salary.

Question 9: Minutes Conversion - Accept total minutes and convert them into hours and remaining minutes using floor division (//) and modulus (%).

Question 10: Seconds Conversion - Accept total seconds and convert them into hours, minutes, and remaining seconds using // and %.

Question 11: Product Bill - Accept product name, quantity, and price per item. Calculate the subtotal, 5% tax, and final payable amount.

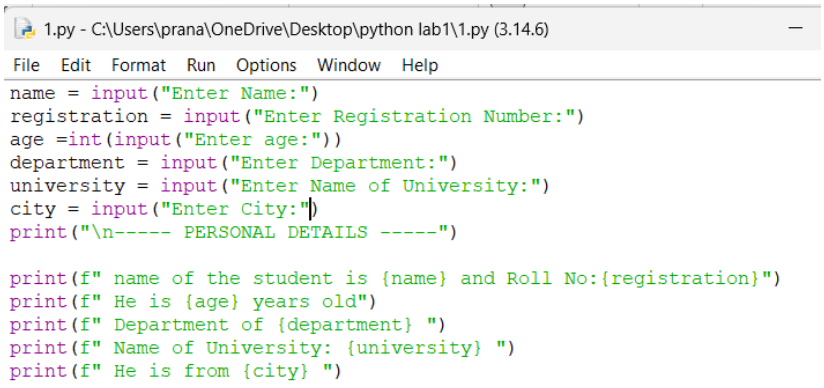
Question 12: Room Area, Perimeter and Total Flooring Cost - Accept the length and breadth of a classroom and the flooring cost per square metre. Display the area, perimeter, and total flooring cost using f-strings with two decimal places.

Programs, Sample Outputs and Results:

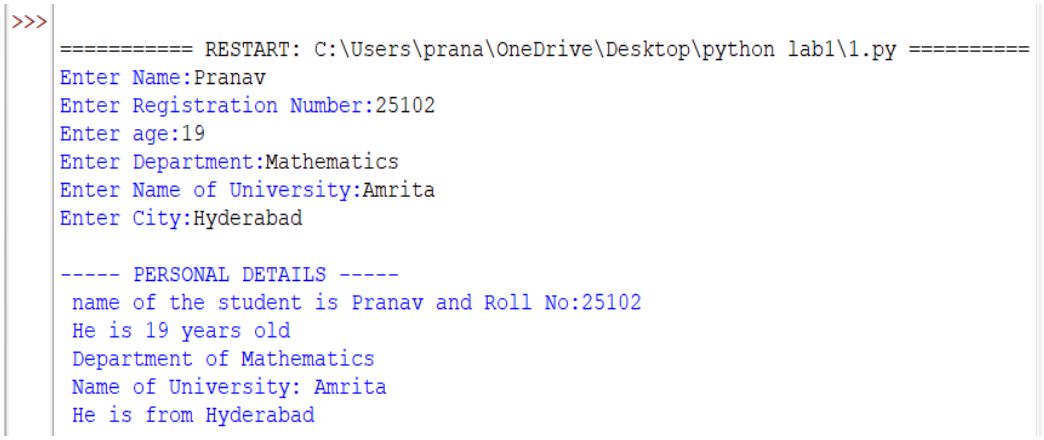
Question 1: Printing Personal Details

Problem Statement: Write a Python program to accept and display the student name, registration number, age, department, university name, and city. Also print one sentence using an f-string.

Program:



Input and Output:



Result: The program accepted and displayed the student details successfully using input(), print(), and an f-string.

Question 2: Sum of Two Numbers

Problem Statement: Accept two integers from the user and display their sum

Program:

2.py - C:\Users\prana\OneDrive\Desktop\python lab1\2.py (3.14.6)

File Edit Format Run Options Window Help

```
a = int(input("Enter First Number:"))
b = int(input("Enter Second Number:"))

s = a + b
print("The Sum of two numbers is ",s)
```

Input&Output:

```
>>>
========== RESTART: C:\Users\prana\OneDrive\Desktop\python lab1\2.py ==========
Enter First Number:2
Enter Second Number:33
The Sum of two numbers is  35
>>>
```

Result: The sum of the two entered integers was calculated and displayed correctly.

Question 3: Basic Arithmetic Operations

Problem Statement: Accept two numbers and display their sum, difference, product, quotient, floor quotient, remainder, and power.

Program:

3.py - C:\Users\prana\OneDrive\Desktop\python lab1\3.py (3.14.6)

File Edit Format Run Options Window Help

```
a = int(input("Enter First Number:"))
b = int(input("Enter Second Number"))
s = a+b
d = a-b
m = a*b
q = a/b
fq = a // b
r = a % b
p = a**b
print(f"The Sum is {s} difference is {d} multiplication {m} quotient {q} floor {fq} remainder {r} power{p}")
```

Input&Output:

```
>>>
========== RESTART: C:\Users\prana\OneDrive\Desktop\python lab1\3.py ==========
Enter First Number:10
Enter Second Number2
The Sum is 12 difference is 8 multiplication 20 quotient 5.0 floor 5 remainder 0 power100
>>>
```

Result: All seven arithmetic operations were performed and displayed correctly.

Question 4: Square and Cube

Problem Statement: Accept one number and calculate its square and cube using the exponentiation operator (**).

Program:

```
a = int(input("Enter Number:"))
s = a**2
c = a**3
print(f"The Square is {s} and Cube is {c}")
```

Input&Output:

```
>>>
========== RESTART: C:\Users\prana\OneDrive\Desktop\python lab1\4.py =========
Enter Number:3
The Square is 9 and Cube is 27
>>>
```

Result: The square and cube of the entered number were calculated using the exponentiation operator.

Question 5: Simple Interest

Problem Statement: Accept principal amount, annual rate of interest, and time period. Calculate the simple interest and total amount.

Program:

5.py - C:\Users\prana\OneDrive\Desktop\python lab1\5.py (3.14.6)

File Edit Format Run Options Window Help

```
p = float(input("Enter Principle Value:"))
r = float(input("Enter Rate:"))
t = float(input("Enter Time value:"))
SimpleInterest = float((p*r*t)/100)
TotalAmount = float(p + SimpleInterest)
print(f"The Interest is {SimpleInterest} and Amount is {TotalAmount}")
```

Input&Output:

```
>>>
========== RESTART: C:\Users\prana\OneDrive\Desktop\python lab1\5.py ==========
Enter Principle Value:10
Enter Rate:10
Enter Time value:10
The Interest is 10.0 and Amount is 20.0
>>>
```

Result: The simple interest and total amount were calculated correctly from the entered values.

Question 6: Area of a Triangle

Problem Statement: Accept the base and height of a triangle and calculate its area.

Program:

6.py - C:\Users\prana\OneDrive\Desktop\python lab1\6.py (3.14.6)

File Edit Format Run Options Window Help

```
base = float(input("Enter base Value of Triangle:"))
height = float(input("Enter height of Triangle:"))
area = 0.5 * base *height
print("The Area of triangle is ",area)
```

Input&Output:

```
>>>
========== RESTART: C:\Users\prana\OneDrive\Desktop\python lab1\6.py ==========
Enter base Value of Triangle:5
Enter height of Triangle:14
The Area of triangle is  35.0
>>>
```

Result: The area of the triangle was calculated successfully using 0.5 x base x height.

Question 7: Kilometre Conversion

Problem Statement: Accept a distance in kilometres and convert it into metres and centimetres.

Program:

7.py - C:\Users\prana\OneDrive\Desktop\python lab1\7.py (3.14.6)

File Edit Format Run Options Window Help

```
km = float(input("Enter KM:"))
m = km * 1000
cm = km * 100000
print("The Meters is",m,"The Centimeters is",cm,sep=',')
```

Input&Output:

```
>>>
========== RESTART: C:\Users\prana\OneDrive\Desktop\python lab1\7.py ========
Enter KM:100
The Meters is,100000.0,The Centimeters is,10000000.0
>>>
```

Result: The entered distance was converted correctly into metres and centimetres.

Question 8: Salary Calculation

Problem Statement: Accept the basic salary, house rent allowance (HRA), and travel allowance. Calculate and display the gross salary.

Program:

8.py - C:\Users\prana\OneDrive\Desktop\python lab1\8.py (3.14.6)

File Edit Format Run Options Window Help

```
basicSalary = float(input("Enter Your Salary:"))
HRA = float(input("Enter Your House Rent:"))
Travel = float(input("Enter Travel Assurance:"))
Total = basicSalary + HRA + Travel
print(f"Salary {basicSalary} and HouseRent {HRA} and Travel {Travel}")
print("The Total Allowance :",Total)
```

Input&Output:

```
>>>
========== RESTART: C:\Users\prana\OneDrive\Desktop\python lab1\8.py ==========
Enter Your Salary:100000
Enter Your House Rent:20000
Enter Travel Assurance:2000
Salary 100000.0 and HouseRent 20000.0 and Travel 2000.0
The Total Allowance : 122000.0
>>>
```

Result: The gross salary was calculated by adding the basic salary, HRA, and travel allowance.

Question 9: Minutes Conversion

Problem Statement: Accept total minutes and convert them into hours and remaining minutes using floor division (//) and modulus (%).

Program:

9.py - C:\Users\prana\OneDrive\Desktop\python lab1\9.py (3.14.6)

File Edit Format Run Options Window Help

```
minutes = float(input("Enter in minuites:"))
hours = minutes // 60
remaining = hours % 60
print(f"The Hours {hours} and Remaining Hours {remaining}")
```

Input&Output:

```
>>>
========== RESTART: C:\Users\prana\OneDrive\Desktop\python lab1\9.py ===
Enter in minuites:60
The Hours 1.0 and Remaining Hours 1.0
>>>
```

Result: The entered minutes were converted correctly into hours and remaining minutes.

Question 10: Seconds Conversion

Problem Statement: Accept total seconds and convert them into hours, minutes, and remaining seconds using // and %.

Program:

10.py - C:\Users\prana\OneDrive\Desktop\python lab1\10.py (3.14.6)

File Edit Format Run Options Window Help

```
total_seconds = int(input("Enter total seconds: "))
hours = total_seconds // 3600
remaining_seconds = total_seconds % 3600
minutes = remaining_seconds // 60
seconds = remaining_seconds % 60
print(f"Hours = {hours}")
print(f"Minutes = {minutes}")
print(f"Seconds = {seconds}")
```

Input&Output:

```
>>>
========== RESTART: C:\Users\prana\OneDrive\Desktop\python lab1\10.py
Enter total seconds: 3600
Hours = 1
Minutes = 0
Seconds = 0
>>>
```

Result: The entered seconds were converted correctly into hours, minutes, and remaining seconds

Question 11: Product Bill

Problem Statement: Accept product name, quantity, and price per item. Calculate the subtotal, 5% tax, and final payable amount.

Program:

11.py - C:\Users\prana\OneDrive\Desktop\python lab1\11.py (3.14.6)

File Edit Format Run Options Window Help

```
p = input("Enter Product Name:")
q = float(input("Enter Quantity of Product:"))
price = float(input("Enter Price:"))
Tax = (price*5)/100
Total = (q*price)+ Tax
print(p)
print(q)
print(price)
print("The Total Amount :",Total)
```

Input&Output:

```
>>>
========== RESTART: C:\Users\prana\OneDrive\Desktop\python lab1\11.py ==========
Enter Product Name:Eggs
Enter Quantity of Product:12
Enter Price:7
Eggs
12.0
7.0
The Total Amount : 84.35
```

Result: The subtotal, tax, and final payable amount were calculated and displayed correctly.

Question 12: Room Area, Perimeter and Total Flooring Cost

Problem Statement: Accept the length and breadth of a classroom and the flooring cost per square metre. Display the area, perimeter, and total flooring cost using f-strings with two decimal places.

Program:

12.py - C:/Users/prana/OneDrive/Desktop/python lab1/12.py (3.14.6)

File Edit Format Run Options Window Help

```
length = float(input("Enter classroom length in metres: "))
breadth = float(input("Enter classroom breadth in metres: "))
cost_per_square_metre = float(
 input("Enter flooring cost per square metre: ")
)
area = length * breadth
perimeter = 2 * (length + breadth)
total_flooring_cost = area * cost_per_square_metre
print("\n----- ROOM CALCULATION -----")
print(f"Area = {area:.2f} square metres")
print(f"Perimeter = {perimeter:.2f} metres")
print(f"Total Flooring Cost = Rs. {total_flooring_cost:.2f}")
```

Input&Output:

```
>>>
========== RESTART: C:/Users/prana/OneDrive/Desktop/python lab1/12.py =========
Enter classroom length in metres: 100
Enter classroom breadth in metres: 100
Enter flooring cost per square metre: 200

----- ROOM CALCULATION -----
Area = 10000.00 square metres
Perimeter = 400.00 metres
Total Flooring Cost = Rs. 2000000.00
>>>
```

Observation:

- The input() function returns data as a string unless it is converted using int() or float().
- Commas in print() can display strings and numerical values together without explicit conversion.
- An f-string begins with the letter f and allows variables or expressions inside curly brackets.
- The / operator performs normal division, // gives the floor quotient, and % gives the remainder.
- The ** operator is used for exponentiation, such as number ** 2 for the square.
- Formatting such as :.2f displays a floating-point value with two decimal places.

Overall Result:

All assigned programs for input, output, type conversion, and arithmetic operations were written and executed successfully in Python IDLE. The outputs were verified for the given sample inputs